# CTMFNet: CNN and Transformer Multiscale Fusion Network of Remote Sensing Urban Scene Imagery

Pengfei Song, Jinjiang Li, Zhiyong An, Hui Fan, and Linwei Fan

***Abstract*— Semantic segmentation of remotely sensed urban scene images is widely demanded in areas such as land cover mapping, urban change detection, and environmental protection. With the development of deep learning, methods based on convolutional neural networks (CNNs) have been dominant due to their powerful ability to represent hierarchical feature information. However, the limitations of the convolution operation itself limit the network's ability to extract global contextual information. With the successful use of transformer in computer vision in recent years, transformer has shown great potential for modeling global contextual information. However, transformer is not sufficiently capable of capturing local detailed information. In this article, to explore the potential of the joint CNN and transformer mechanism for semantic segmentation of remotely sensed urban scenes, we propose a CNN and transformer multiscale fusion network (CTMFNet) based on encoding–decoding for urban scene understanding. To couple local–global context information more efficiently, we designed a dual backbone attention fusion module (DAFM) to couple the local and global context information of the dual-branch encoder. In addition, to bridge the semantic gap between scales, we built a multilayer dense connectivity network (MDCN) as our decoder. The MDCN enables the full flow of semantic information between multiple scales to be fused with each other through upsampling and residual connectivity. We conducted extensive subjective and objective comparison experiments and ablation experiments on both the International Society of Photogrammetry and Remote Sensing (ISPRS) Vaihingen and ISPRS Potsdam datasets. Numerous experimental results have proven the superiority of our method compared to currently popular methods.**

***Index Terms*— Convolutional neural network (CNN), multiscale fusion, semantic segmentation, transformer.**

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## I. Introduction

In RECENT years, with the rapid development of sensor technology and artificial intelligence, the semantic segmentation of high-resolution remotely sensed urban scene images has become a hot topic in the field of remote sensing images. Remote sensing images play a crucially important role in various urban applications, such as urban planning [1], vehicle detection [2], land cover mapping [3], time series change detection [4], and building and road extraction [5], [6], among other application scenarios. Semantic segmentation is a pixel-level classification of the whole image, labeling each pixel with the appropriate category (e.g., building, road, tree, vehicle, etc.). However, as objects in urban scenes are characterized by large intra-class differences and small inter-class differences. Therefore, semantic segmentation for high-resolution remote sensing images remains a challenging task. In recent years, with the rapid development of convolutional neural networks (CNNs), many researchers have proposed several CNN-based methods for solving the problem of semantic segmentation of city scenes in high-resolution remote sensing images. In particular, the full convolutional network (FCN) [7] based approach allows for end-to-end training and truly enables segmentation at the pixel level. Compared to traditional approaches to machine learning such as support vector machines (SVMs) [8], random forests [9], and conditional random fields (CRF) [10], deep convolutional network-based methods can capture more detailed local feature information and have powerful local feature extraction capabilities. This gives it a huge advantage in feature representation and pattern recognition.

Although FCN achieves good results, it lacks the use of multiscale feature information due to the oversimplified design of its decoder section. Subsequently, the decoder–encoder structure, represented by UNet [11], further improves accuracy using a skip connection to make up for the feature information lost by downsampling. However, the low-level features generated by the encoder are linked to the high-level features generated by sampling on the decoder without further refinement, which does not make sufficient use of the features and leaves the network with insufficient ability to discriminate between them. At the same time, the presence of complex backgrounds and large amounts of noise in remotely sensed images, along with the need to segment many categories, such as buildings, vehicles, roads, and other categories, requires us

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Fig. 1. Illustration of local and global contextual information.

to pay more attention to the semantic information in multiscale features.

Contextual fusion using multiple scales is a viable solution [12], [13], [14] to enhance semantic interactions and enrich feature semantic information. In general, to deepen the multiscale feature representation of the network without adding significant computational complexity, common approaches to multiscale semantic feature processing mainly include multilevel feature fusion (MFF) and spatial pyramid pooling (SPP) strategies. MFF in general establishes the relationship between low-level features and high-level features by residual connection or skip connection, and the merged features are gradually restored to the resolution of the original image [15], [16]. SPP uses dilated convolution to extend the convolution's perceptual field and capture multiscale feature information [17], [18]. Analyzing previous approaches to multiscale fusion, MFF usually establishes unidirectional (bottom-up or left-to-right) multiscale interactions, which do not make sufficient use of the multiscale feature information extracted by the encoder. Second, the dilated convolution in SPP is not friendly to small target objects, which can easily lose their spatial information and cause grid effects.

In addition, due to the limitations of convolution itself, specifically, convolution operations are designed to extract local features and lack the ability to model global contextual information. In CNN, each convolutional kernel focuses only on the local set of pixels in its perceptual field and cannot model long-distance dependencies. Although attention mechanisms can be combined with CNN-based structures to establish long-range dependencies of features [19], [20], they are still not liberated from CNNs. Recently, with the successful use of the visual transformer, additional solutions have been brought to this end [21], [22], [23]. The transformer uses a multi-headed self-attentive mechanism to capture the global feature information of an image to create effective long-range dependencies, avoiding the need for convolution operations altogether. Although transformer captures long-range dependencies of global features well, it pays less attention to local feature information. Therefore, it is crucial to establish a mechanism to unite the local semantic feature information from CNN with the global contextual features from transformer [24].

Based on the above considerations, in this article we propose an united CNN and transformer multiscale fusion network (CTMFNet), based on the traditional decoding–encoding structure. CTMFNet consists of a dual backbone branch of CNN and transformer to extract local and global feature information, and a dual backbone attention fusion module (DAFM) to realize the coupling of local and global feature information, as shown in Fig. 1. In addition, to enhance the multiscale semantic relationships for accurate segmentation, we build multi-layer dense connectivity networks (MDCNs) as decoders based on DenseNet [25]. In MDCN, left-to-right feature reuse is achieved based on the structure of DenseNet itself, and a bottom-up multiscale feature fusion is established by upsampling. Both enhance feature interaction at all stages in a comprehensive manner, making fuller use of multiscale feature information.

Our work contributions can be summarized as follows.

1) We propose a novel joint CTMFNet, to achieve accurate semantic segmentation. CTMFNet uses CNN and transformer as the dual-branch backbone to extract local and global feature information. The dual-branch backbone information is effectively coupled by the DAFM. The coupled multiscale feature information enhances the interaction between features through MDCNs to achieve bidirectional multiscale feature fusion, thus improving the accuracy of segmentation.
2) To effectively couple the coding features from CNN and transformer, we designed the DAFM. DAFM contains a self-attention mechanism and a cross-fusion mechanisms. Local and global features are fused, after which they further go through a self-attentive mechanism to focus on focal area information.
3) To achieve efficient multiscale feature interaction, we adopt a MDCN as our decoder, allowing sufficient flow of semantic information. The residual connections are used to create a left-to-right transfer of features, making more efficient use of features. Connecting features between different levels by upsampling to create bottom-up information interactions.

The main research in this article is as follows. In Section II, we present the current state of research on semantic segmentation of high-resolution remote sensing images. We will describe the methods we use in detail in Section III. In Section IV, we will perform subjective and objective comparison experiments with current popular methods and ablation experiments to verify the robustness of our network. Section V is a summary.

## II. Related Work

### A. CNN-Based Remote Sensing Image Segmentation

The FCNs [7] are the first approach to achieving effective semantic segmentation in an end-to-end manner. Since then, methods based on FCN have dominated the field of semantic segmentation in remote sensing [25], [26]. However, the

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accuracy of the final segmentation is compromised by the FCN's rudimentary decoder design. Subsequently, UNet [11] based on the encoder–decoder structure was proposed to solve this problem. The encoder stage extracts multiscale features by continuously reducing the spatial resolution of the features and increasing the channel dimension. The decoder stage recovers the spatial resolution layer by layer and uses skip connections to merge the encoder and decoder features, reducing the impact caused by the reduced spatial resolution in the encoding stage. Subsequently, the encoder–decoder network framework became the standard framework for semantic segmentation of remote sensing images [27]. In the encoder–decoder structure, to capture richer contextual features and reduce the loss of feature information, ResUnet-a [28] combines residual concatenation, atrous convolution, pyramidal scene understanding, and multi-task inference to establish conditional relations between various tasks and improve the accuracy of segmentation. Although the decoder–encoder-based CNN approach has achieved satisfactory results. However, due to the presence of complex man-made buildings and target objects of varying scales in high-resolution remotely sensed urban images, the CNN-based approach, with its limited field of perception, lacks the ability to model the global information of the whole image. It is difficult to accurately identify these complex objects if we only rely on local feature information.

To address the problem of limited perceptual fields in CNNs, researchers have incorporated attentional mechanisms into the network structure to model global feature information. Multistage attention residual Unet. (MAResU-Net) [29] reconstructs the skip connections in the original UNet by adding a linear attention mechanism (LAM) to each skip connection to establish the long-term dependencies of the feature graph. MACU-Net [19] redesigned a multiscale skip connection combination and re-alignment of the semantic features contained in the low-level and high-level feature maps on the basis of ResUnet. Multi attention network (MANet) [12] extracts contextual dependencies to establish long-term dependencies of the feature graph through multiple efficient attention modules. Also to alleviate the large computational demands in attention, it proposes a new kernel attention with linear complexity. Attentive bilateral contextual network (ABCNet) [20] uses bilateral paths, spatial paths that retain rich spatial detail and contextual paths that capture global contextual information, to fuse features using the feature aggregation module (FAM). $A^2$-FPN [30] uses feature pyramids to encode semantic features at multiple scales, and augments multiscale feature learning with an attention aggregation module (AAM). However, the attention module of the above methods either uses a dot product mechanism or a linear mechanism, relying excessively on convolution operations and thus limiting the representation of global features.

### *B. Semantic Segmentation of Remote Sensing Images Based on Transformer*

Recently, with the rise of transformer in the field of computer vision, it has received increasing attention from research scholars. Due to transformer's powerful sequence-to-sequence modeling capabilities, it exhibits superior performance in global context modeling than the separate attention models described above. Most of the existing approaches to semantic segmentation also continue to follow the encoder–decoder framework. These approaches can be broadly classified into two categories: the first is an encoder–decoder architecture consisting entirely of transformers, such as segmenter [31], SegFormer [32], and SwinUNet [33]. The second category is a hybrid architecture, with either CNN or transformer as the encoder and the other as the decoder. Transformer-based semantic segmentation of remote sensing images is usually the second structure to explore the effective joint mechanism of CNN and transformer. Bilateral awareness network (BANet) [23] constructs two feature extraction paths: one path uses CNN to extract local texture features and the other path uses transformer to capture long-range dependencies. A FAM is also designed to fuse the extracted texture and global features from the two backbone paths. Densely connected swin (DC-Swin) [22] uses the transformer as a feature extractor for extracting multiscale global contextual feature information. The global multiscale features are fused by a densely connected feature aggregation module (DCFM) designed to extract multiscale semantic features enhanced by semantic relationships.

## III. Methodology

In this section, we will elaborate on the details of the architecture and component modules of CTMFNet. The general structure of the network is described first, followed by the dual branch backbone path, the DAFM and the MDCN.

### *A. Overall Structure*

The overall network structure of the CTMFNet we have designed is shown in Fig. 2. In our CTMFNet structure, we use high-resolution network (HRNet) [34] as our CNN backbone path, and CrossFormer [21] as our transformer backbone path to extract multiscale local texture feature information and global contextual feature information. And then, the local–global feature information is fused by our designed DAFM to establish an efficient CNN-transformer joint mechanism. The fused features enable bidirectional multiscale feature exchange through our MDCN, improving the exchange of information between layers and allowing semantic information to flow fully in the decoder.

Suppose, our input red, green, blue (RGB) image is $F \in \mathbb{R}^{3\times H\times W}$. The local and global feature information $f_i^L \in \mathbb{R}^{C_i\times(H/2^i)\times(W/2^i)}$, $i \in \{1, 2, 3, 4\}$, $C_i \in \{48, 96, 192, 384\}$, and $f_i^G \in \mathbb{R}^{C_i\times(H/2^i)\times(W/2^i)}$, $C_i \in \{64, 128, 256, 512\}$ are extracted by CNN with transformer dual branch backbone. We then feed the four-layer multiscale local texture features $f_i^L$ and the global context features $f_i^G$ into DAFM to effectively couple the local texture features with the global context features. Self-attentive mechanisms and cross-fusion mechanisms are included in DAFM. We first cross-fuse the input local texture features $f_i^L$ with the global contextual features $f_i^G$ by convolutional units separately, and then further fuse the fused dual-branch features. The fused features are further subjected to a pyramid split attention (PSA) mechanism, which allows the network to selectively emphasize interdependent feature maps

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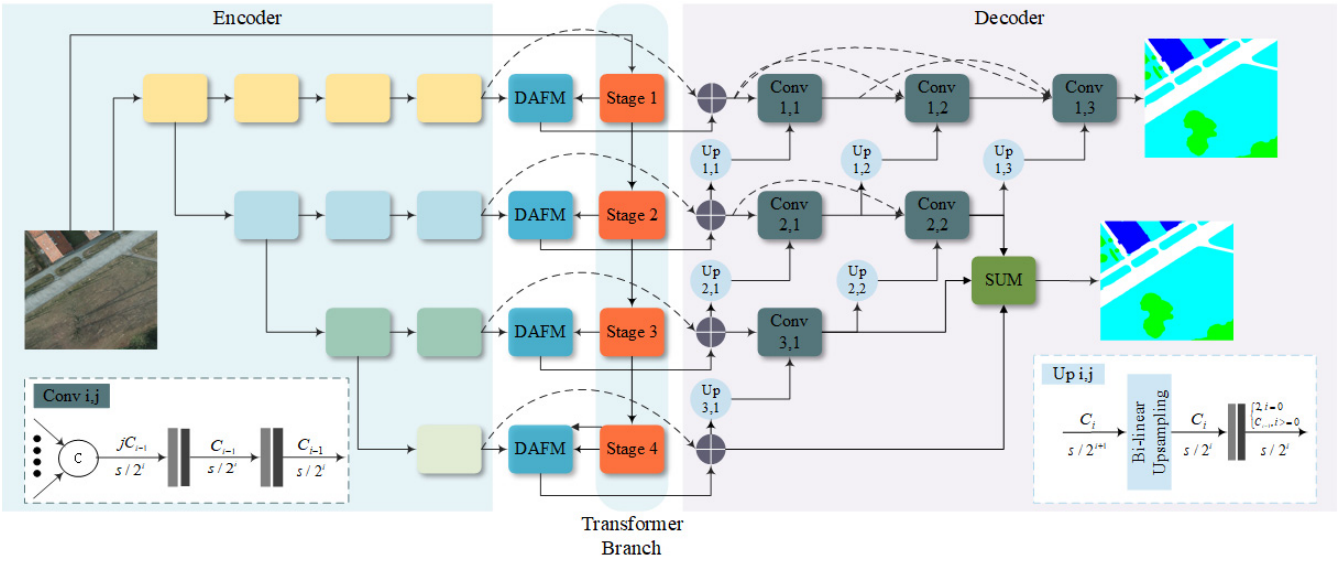



Fig. 2. CTMFNet network structure. It mainly consists of a feature extractor consisting of CNN and transformer, a dual branch attention fusion module, and a multilayer densely connected subnetwork as decoder.

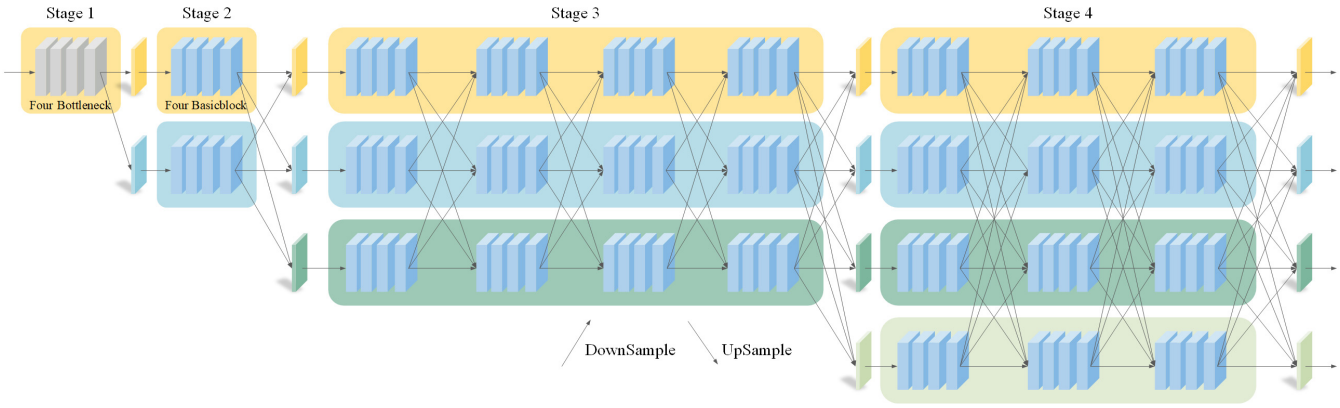



Fig. 3. HRNet network structure, consisting of a sequentially increasing four-stage subnetwork.

from a global perspective and to be more sensitive to each category. Afterward, we transfer the fused feature map $f_i^D$ to a MDCN using up-sampling to recover spatial resolution layer by layer, building a bottom-up, left-to-right multiscale feature exchange to generate the final segmentation prediction map $P_1$. Finally, to better optimize our MDCN, we additionally built an auxiliary head sum (SUM) to generate a secondary segmentation prediction map $P_2$ to assist in optimizing the MDCN and to better optimize our overall network structure.

Our overall network structure is defined as follows, $i \in [1, 4]$:

$$f_i^L = \text{HRNet}(F) \tag{1}$$
$$f_i^G = \text{CrossFormer}(F) \tag{2}$$
$$f_i^D = \text{DAFM}(f_i^L, f_i^G) \tag{3}$$
$$P_1 = \text{MDCN}(f_i^D). \tag{4}$$

### B. Dual Branch Backbone

*1) CNN Branch:* In traditional semantic segmentation of remote sensing images, the downsampling performed at the encoding stage makes it difficult to recover fine edge information by upsampling at the decoding stage due to the high resolution of remote sensing images. To minimize the loss of information caused by the reduced spatial resolution of high-resolution remote sensing images, we use the HRNet as our encoder. The HRNet always maintains feature information in high-resolution by connecting high-resolution and low-resolution convolutional features in parallel. Each stage receives repeated information from the other parallel stages, resulting in a richer representation of high-resolution features.

In the experiments, HRNet contains four stages and four layers of subnetworks with different feature scales. The start of each stage is made by the upper layer generating a new feature representation by downsampling to generate a new layer of the subnetwork. This is shown in Fig. 3. The $N$ stage contains the $N$ layer subnetwork. Each stage contains 1, 1, 4, 3 convolution modules in turn, and a total of eight multiscale fusions were performed. The first stage of the convolution module is similar to the bottleneck block of residual net-50 (ResNet-50). The remaining stages are basicblock convolution blocks, each containing two $3 \times 3$ convolution blocks and

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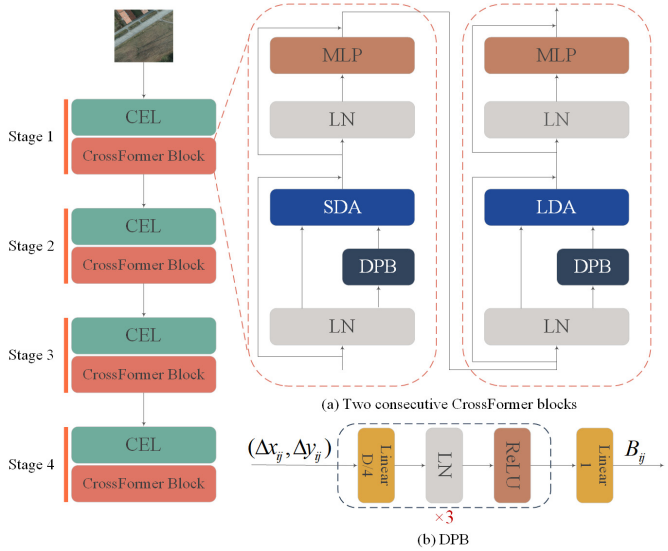



Fig. 4. CrossFormer network structure, consisting of cross-scale embedding layer and CrossFormer block. In CrossFormer block, cross-scale attention computation is implemented by SDA and LDA. (a) Two consecutive CrossFormer blocks. (b) DPB.

upsampling blocks for multiscale fusion. The size of the feature maps of the four parallel branches is in the order of $1/4, 1/8, 1/16, 1/32$ of the original spatial resolution of the input. When fusing features at each scale, the high-resolution feature map is downsampled and the low-resolution feature map is upsampled. The features are replicated at the same resolution to ensure that the high resolution has a rich feature representation.

*2) Transformer Branch:* Compared to the currently popular vision transformer (ViT) [35], [36], CrossFormer places more emphasis on the ability to build cross-scale features and create cross-scale attention between features at different scales. CrossFormer uses a pyramidal structure, thus dividing the model into four stages, as shown in Fig. 4. As can be seen in Fig. 4, CrossFormer consists of two core modules, cross-scale embedding layer (CEL), and CrossFormer block [containing long short distance attention (LSDA), and dynamic position bias (DPB)].

*Cross-Scale Embedding Layer:* The CEL is located at the beginning of each stage, and based on the output of the previous stage or the original input image as input. Several different Kernel convolutional layers are used to sample the input, resulting in several feature maps of different sizes. The feature maps of different scales are then mapped and stitched together to produce an embedding. Each embedding thus contains feature information at a different scale of the corresponding region. Also, taking into account issues such as computational overhead, larger kernels are allocated to smaller dimensions and smaller kernels to larger dimensions.

*CrossFormer Block:* Each CrossFormer block contains a LSDA and a multilayer perceptron (MLP). This is shown in Fig. 4(a). The two CrossFormer blocks alternate between short distance attention (SDA) and long distance attention (LDA) for cross-scale attention computation. DPB is added to obtain the location representation of the embedding. Every two consecutive CrossFormer blocks are considered to be a self-attentive group.

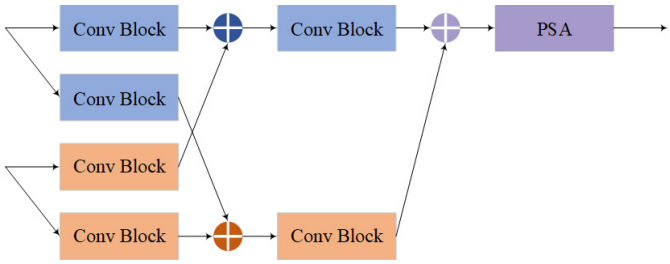



Fig. 5. Structure diagram of the DAFM.

*Long Short Distance Attention:* Previous self-attentive mechanisms had too much computational overhead for visual inputs. Therefore, in order not to destroy the fine-grained features of the embedded features, self-attentiveness is divided into SDA and LDA. SDA and LDA use the same attention calculation as ViT, with the difference that the embedding inputs for each layer are divided into many groups and the calculation of self-attention is restricted to within each group. The main difference between SDA and LDA is that the embedding inputs are grouped differently. SDA divides similar embeddings into groups of size $G$. LDA groups are sampled at intervals $I$ and each position is sampled once at an interval of $I$ positions. SDA is responsible for constructing dependencies between adjacent embeddings, while LDA is responsible for dependencies between long-range embeddings. In this way, LSDA reduces the cost of self-attentiveness without compromising small- or large-scale features and helps multiscale feature interaction.

*Dynamic Position Bias:* To be more flexible in adapting to inputs of different sizes, a trainable DPB is proposed that takes the distance of two embeddings as input and outputs their position bias. This is illustrated in Fig. 4(b). The nonlinear variation of the DPB consists of three fully connected layers with normalization. The input to the DPB is $(\Delta x_{ij}, \Delta y_{ij})$ denotes the coordinate distance between the $i$th and $j$th embedded features, and the dimension of the middle linear layer is $D/4$, where $D$ denotes the dimension of the embedding. The output $B_{ij}$ is a scalar representing the relative position features between the embedded features

$$B_{ij} = \mathrm{DPB}(\Delta x_{ij}, \Delta y_{ij}). \tag{5}$$

### C. Dual Branch Attention Fusion Module

To more effectively couple the multiscale coding features from the CNN and transformer dual-branch backbones, we designed a new DAFM, as shown in Fig. 5. It contains both self-attentive mechanisms and cross-fusion mechanisms. Specifically, when our DAFM receives multiscale local texture features $f_i^L$ from the CNN and multiscale global contextual features $f_i^G$ from the transformer. First, we map two new feature tensors $f_i^{L`}$ and $f_i^{G`}$ using a $3 \times 3$ convolution unit. The number of channels $f_i^{L`}$ and $f_i^{G}$ are the same, as are channels $f_i^{G`}$ and $f_i^{L}$. And then $f_i^{L`}$ and $f_i^{G}$, $f_i^{G`}$ and $f_i^{L}$ are crossed and added to create $\widehat{f_i^L}$ and $\widehat{f_i^G}$, respectively. The summed features are again passed through a $3 \times 3$ convolution

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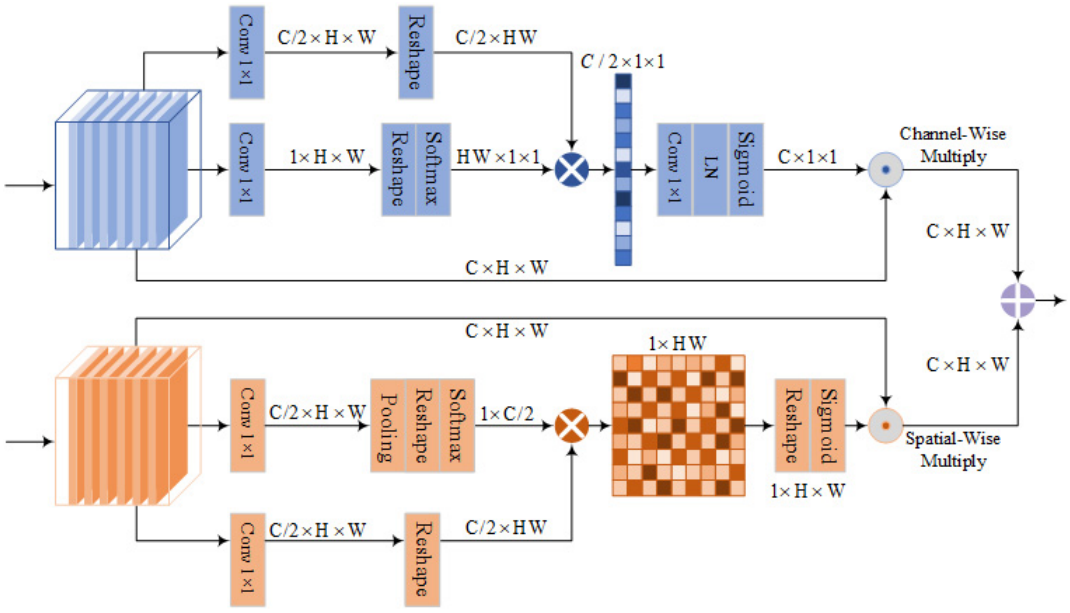



Fig. 6. Structure diagram of PSA attention. It contains two branches, the spatial attention branch, and the channel attention branch.

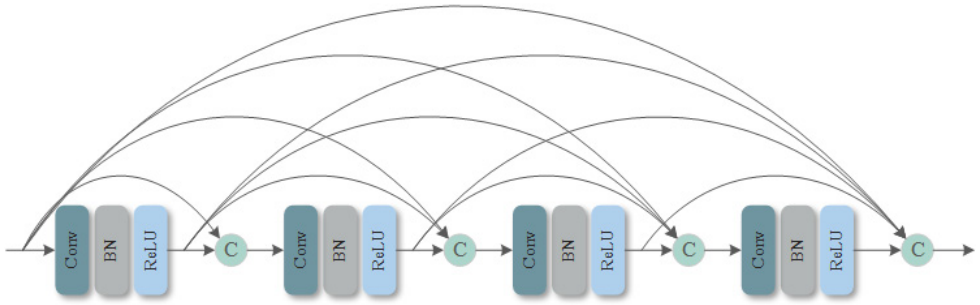



Fig. 7. Schematic of a densely connected network.

unit so that the feature channels of the two branches are the same, and then fused. This allows local and global features to be coupled more effectively, thus creating an effective joint mechanism. The fused feature tensor is further subjected to PSA to enhance the detailed features of the feature regions and suppress information interference from irrelevant regions, enriching the feature representation of the target region. The PSA structure is shown in Fig. 6. The coupling operation of the DAFM module can be expressed as follows:

$$\widehat{f_i^L} = \text{Conv}(f_i^L) + \text{Conv}(f_i^G) \tag{6}$$

$$\widehat{f_i^G} = \text{Conv}(f_i^G) + \text{Conv}(f_i^L) \tag{7}$$

$$f_i^D = \text{PSA}(\widehat{f_i^L} + \widehat{f_i^G}). \tag{8}$$

### *D. Multi-Layer Densely Connected Sub-Networks*

Existing studies [12], [13], [14] have shown that establishing an effective multiscale fusion mechanism is essential to accurately identify the target region and improve the accuracy of segmentation. To achieve efficient multiscale feature fusion and maximize the full flow of feature information between different scales, based on DenseNet, we build a MDCN as our decoder for recovering spatial resolution layer by layer. As shown in Fig. 2. The MDCN adds convolutional modules in order from bottom to top $\text{Conv}_{i,j}$, where $i$ indicates the first layer of the subnetwork and $j$ indicates the first stage. Each convolutional module has two convolutional units with a convolutional kernel size of $3 \times 3$, BatchNorm2d, and rectified linear unit (ReLU). The resolution is recovered layer by layer by upsampling, $Up_{i,j}$, high-level features between layers, where $i$ denotes up-sampling to the first layer of the subnetwork and $j$ denotes the low-level stage. Each upsampling module consists of a convolutional unit with a convolutional kernel size of $3 \times 3$ and a bilinear upsampling function. Establish bottom-up information exchange between layers through up-sampling to achieve multiscale feature fusion.

A densely connected network with a single layer is shown in Fig. 7. Intuitively, a single-layer densely connected network may require a higher number of parameters. However, there are not many network parameters required relative to ResNet, as each layer processes the feature map simply by matrix stitching to achieve residual connectivity. This enables feature reuse in the channel dimension and improves the flow of information in the decoder as well as gradient changes, allowing

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each layer to be directly linked to the input feature tensor and loss function, thus mitigating gradient disappearance.

In terms of network structure, it corresponds to the CNN backbone branch of the decoder, also divided into four layers of subnetworks and four stages. The first layer of the subnetwork contains three convolutional modules. The remaining subnetworks decrease convolutional modules in turn. There are three upsampling units between the first and second layers, and decreasingly so thereafter. When the coupled feature tensor is received by each layer of the network except the last layer of the decoding network, the two are first stitched together in the channel dimension. The stitched features are passed through the first convolutional unit to extract features and reduce the number of channels. It then undergoes a normalization process, and the processed features are further passed through a second convolutional unit to further smooth the features. The features after the first convolutional unit are first transferred directly to all subsequent convolutional units (if present). Second, they are transmitted through the upsampling module to the upper convolutional unit corresponding to that stage, establishing a bottom-up exchange of information and enabling multiscale feature fusion. When the other convolutional units receive the transmitted feature tensor, a similar operation is performed. If there are $N$ convolutional units in the current layer of the subnetwork, $N$ convolutional operations are performed and the features after the operations are upsampled $N$ times. The first convolution unit receives features that jump backward $N$ times and the output features need to jump backward $N-1$ times. The fourth layer of the subnetwork does not require convolutional operations and is upsampled directly.

In this way, while recovering the spatial resolution layer by layer through the MDCN decoder, it enables the full flow of multiscale features throughout the decoder, achieving efficient multiscale feature fusion. At the same time, MDCN minimizes the loss of spatial information, allowing the network to identify and localize target regions more accurately and improve the accuracy of segmentation.

### *E. Loss Functions*

Based on the studies in [37] and [38], we not only perform supervised optimization of the final segmentation prediction map $P_1$ but also add additional auxiliary headers to aggregate the deep semantic information of the MDCN to generate secondary prediction segmentation maps $P_2$ to optimize the MDCN, as shown in Fig. 2. Based on this design, we train our overall network structure with a primary loss $L_M$ and a secondary loss $L_A$. $L_M$ is a combination of dice loss ($L_D$) and cross-entropy loss ($L_{\mathrm{CE}}$). The primary loss is defined by the following:

$$L_D = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K}\frac{p^{`n}_k p^n_k}{p^{`n}_k + p^n_k} \tag{9}$$

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} p^n_k \log p^{`n}_k \tag{10}$$

$$L_M = L_D + L_{\mathrm{CE}} \tag{11}$$

where $N$ and $K$ denote the number of training samples and the number of categories. $p^{`n}$ and $p^n$ denote the one-hot encoding of ground-truth and the predicted output of our network, respectively. $p^{`n}_k$ denotes the confidence of category $K$ in sample $N$. In addition, the $L_{\mathrm{CE}}$ was chosen as our auxiliary loss ($L_A$), which was used to constrain our secondary segmentation prediction map and optimize our MDCN. The SUM takes as input the MDCN inverse three layers of high-level semantic features and aggregates the three layers by upsampling a $3 \times 3$ convolutional layer with normalization and ReLU, and a $1 \times 1$ convolutional layer to generate a secondary segmentation prediction map. To better combine the primary and secondary losses, we further multiply the secondary losses by the factor $\alpha$. In our experiments, we set $\alpha$ to 0.4. So that our overall loss function $L_{\mathrm{total}}$ is defined as follows:

$$L_{\mathrm{total}} = L_M + \alpha L_A. \tag{12}$$

The algorithmic inference process of our overall framework, as shown in Algorithm 1.

**Algorithm 1** Implementation Process of Our CTMFNet Model

**Input:** $A$ (Original image)
**Output:** $P_1$ (a prediction mask), $P_2$ (a subprediction mask)
// Step1: Extraction of dual features
$\mathbf{X_i} = CNN(\mathbf{A}), \mathbf{Y_i} = Transformer(\mathbf{A})$
// Step2: Dual-branch feature coupling
**for** $i\ in\ \{1,4\}$ **do**
$\mathbf{D_i} = DAFM(\mathbf{X_i}, \mathbf{Y_i})$
**end**
// Step3: Use MDCN to generate prediction mask
**for** $i\ in\ \{1,4\}$ **do**
$\mathbf{M_{5-i}} = MDCN(\mathbf{D_{5-i}})$
**end**
// Prediction mask $\mathbf{P_1} = \mathbf{M_1}$
// Step4: Use SUM to generate subprediction mask
$\mathbf{P_2} = SUM(\mathbf{M_4}, \mathbf{M_3}, \mathbf{M_2})$
// Step5: update the network parameters
The total Loss Funtion $L_{total} = L_M(P_1) + 0.4L_A(P_2)$
The AdamW optimizer and the total loss $L_{total}$ to update the network parameters
**end**

## IV. Experiments and Analysis

To evaluate the performance of our proposed method, we tested by comparing the results of our model with other six currently popular models on datasets International Society of Photogrammetry and Remote Sensing (ISPRS) Vaihingen and ISPRS Potsdam. We also designed an ablation experiment to evaluate the effect of our proposed the transformer branch, DAFM, and MDCN. Finally, we conducted a comprehensive analysis of the experimental results.

### *A. Datasets*

We have demonstrated the superiority of our method by comparing our method with six currently popular methods on two popular datasets: ISPRS Vaihingen, ISPRS Potsdam (http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html).

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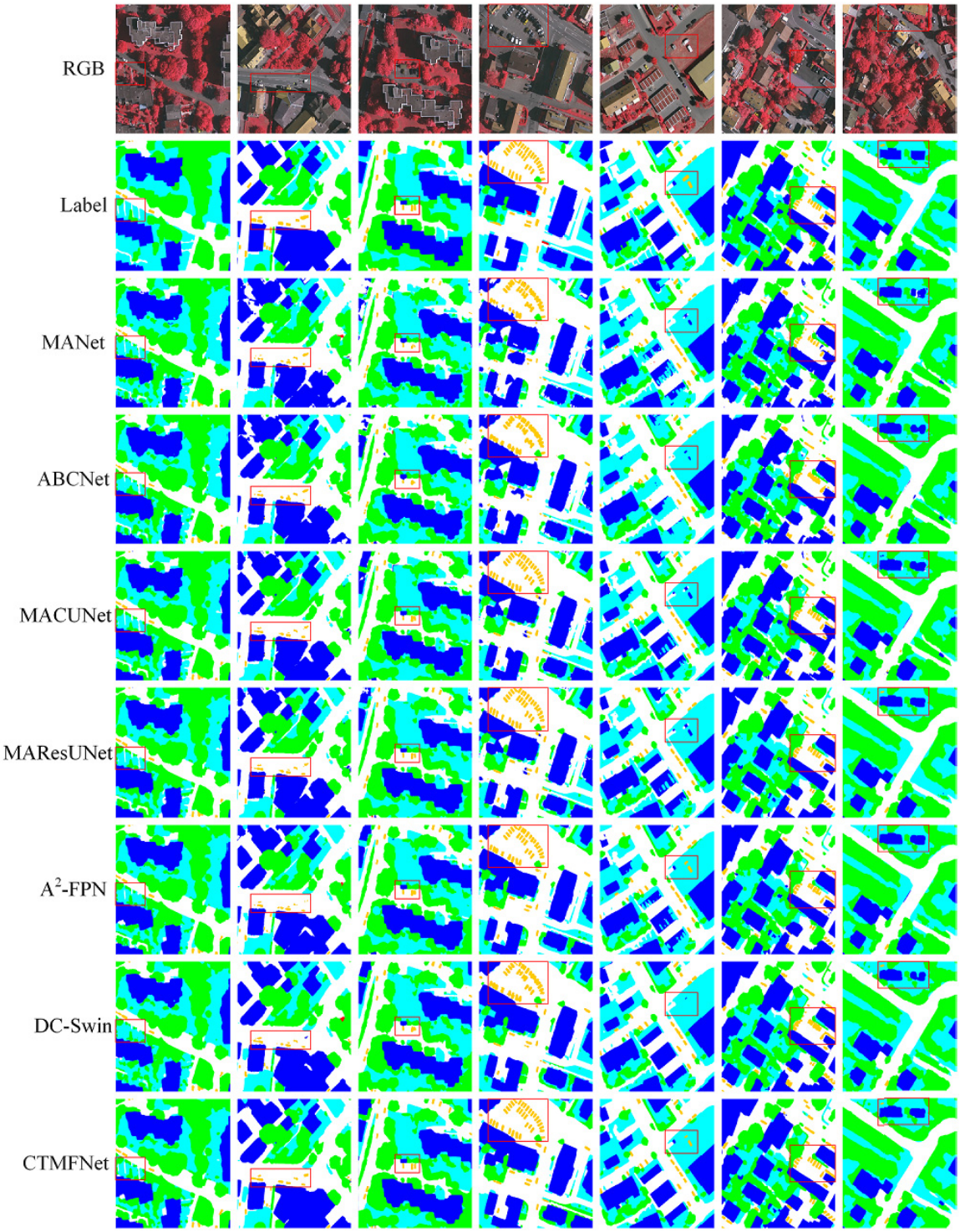



Fig. 8. Qualitative comparisons between our method and other models for the Vaihingen test dataset on patches images.

*1) ISPRS Vaihingen:* It contains 33 highly detailed spatial resolution top (TOP) images (ground sampling distance (GSD) of 9 cm) with an image resolution of 2494 × 2064. Each TOP image contains three spectral images of red, green, and blue together with digital surface models (DSMs) and normalized DSMs (NDSMs). It contains six feature types: impervious surfaces, building, low vegetation, tree, car, and clutter/background. Based on studies [20], [30], we selected IDs: 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, 38 for testing, ID: 30 for validation and the other 15 images for training. In our experiments, we selected RGB images for training and filled and cropped the original input images into patches of 1024 × 1024 pixels in size.

*2) ISPRS Potsdam:* The ISPRS Potsdam dataset contains 38 highly detailed spatially resolved TOP images (GSD of 5 cm) with an image resolution of 6000 × 6000. The dataset

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TABLE I

QUANTITATIVE COMPARISON OF RESULTS ON THE ISPRS VAIHINGEN TEST SET WITH STATE-OF-THE-ART MODELS. THE BEST VALUES IN THE COLUMN ARE IN BOLD. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%). INDICATORS FOR THE CATEGORIES ARE F1 SCORES

| Method | Imp.surf. | Building | Lowveg. | Tree | Car | MeanF1 | mIoU | OA |
|---|---|---|---|---|---|---|---|---|
| MANet [12] | 84.95 | 88.41 | 78.16 | 88.37 | 70.47 | 82.07 | 70.16 | 84.80 |
| ABCNet [20] | 88.13 | 90.23 | 76.71 | 87.21 | 68.72 | 82.20 | 70.58 | 85.62 |
| MACU-Net [19] | 90.70 | 92.40 | 81.90 | 89.37 | 82.53 | 87.38 | 77.85 | 88.61 |
| MAResU-Net [29] | 92.91 | 95.26 | 84.95 | 89.94 | 88.33 | 90.28 | 83.30 | 90.86 |
| $A^2$-FPN [30] | 92.99 | 95.53 | 84.67 | 90.34 | 87.62 | 90.23 | 82.42 | 91.04 |
| DC-Swin [22] | 93.46 | 96.00 | **85.32** | 90.03 | 84.88 | 89.94 | 82.01 | 91.29 |
| CTMFNet | **93.79** | **96.12** | 85.02 | **90.47** | **91.47** | **91.37** | **84.34** | **91.60** |

provides images of multiple spectra (RGB, and infrared) as well as DSM and NDSM. As with the Vaihingen dataset, the same six categories of impervious surfaces, building, low vegetation, tree, car, and clutter/background are included. Based on studies [20], [30], we selected IDs: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, 7_13 for testing, ID: 2_10 for validation, and the remaining 22 images (except for the mislabelled image 7_10) for training. In our experiments, RGB images were selected for training and the original input image was filled and cropped to a patch of 1024 × 1024 pixels.

### *B. Experimental Design and Validation Metrics*

Both our method and current popular methods were trained and validated in the Pytorch 1.8.1 framework on a single NVIDIA TITAN RTX GPU to ensure a fair comparison. To achieve fast convergence during training, we use the AdamW optimizer for training. The initial learning rate is set to 1e-4 and a cosine strategy is used to adjust our learning rate. During the training process, data enhancement techniques such as random vertical flip, random horizontal flip, and random luminance are used. At the same time, all methods involved in the experiment are trained and validated under the same data benchmark to ensure the fairness of the experiment. We set the training epoch to 200 to find the best model parameters.

To more effectively assess the superiority of our method, we selected overall accuracy (OA), the mean intersection over union (mIoU), and the Mean F1 score (F1) to compare with currently popular methods. Based on the accumulated confusion matrix, OA, Mean F1 and mIoU are calculated as follows:

$$\mathrm{OA} = \frac{\sum_{C=1}^{C} TP_C}{\sum_{C=1}^{C} TP_C + FP_C + TN_C + FN_C} \tag{13}$$

$$\mathrm{mIOU} = \frac{1}{C}\sum_{C=1}^{C} \frac{TP_C}{TP_C + FP_C + FN_C} \tag{14}$$

$$\mathrm{F1}_C = 2 \times \frac{\mathrm{Recall}_C \times \mathrm{Precision}_C}{\mathrm{Recall}_C + \mathrm{Precision}_C} \tag{15}$$

where $\mathrm{TP}_C$, $\mathrm{FP}_C$, $\mathrm{TN}_C$, $\mathrm{FN}_C$ represent true positive, false positive, true negative, and false negatives when the category is $C$ respectively

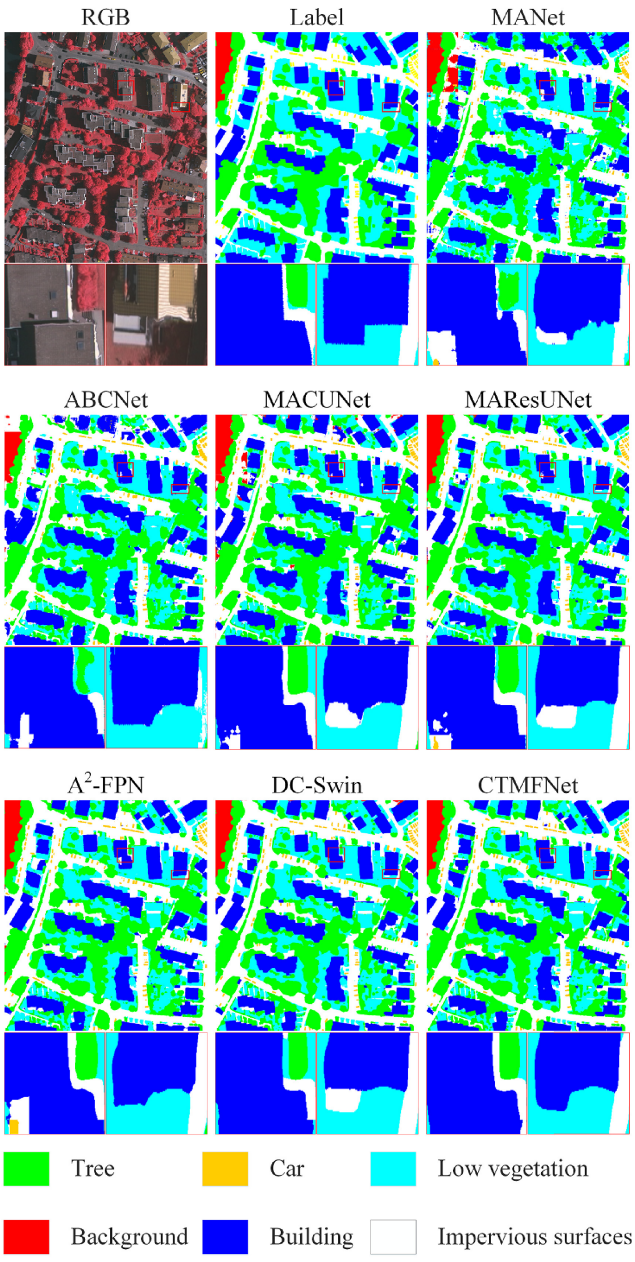



Fig. 9. Qualitative comparisons between our method and other models for the Vaihingen test dataset on whole images.

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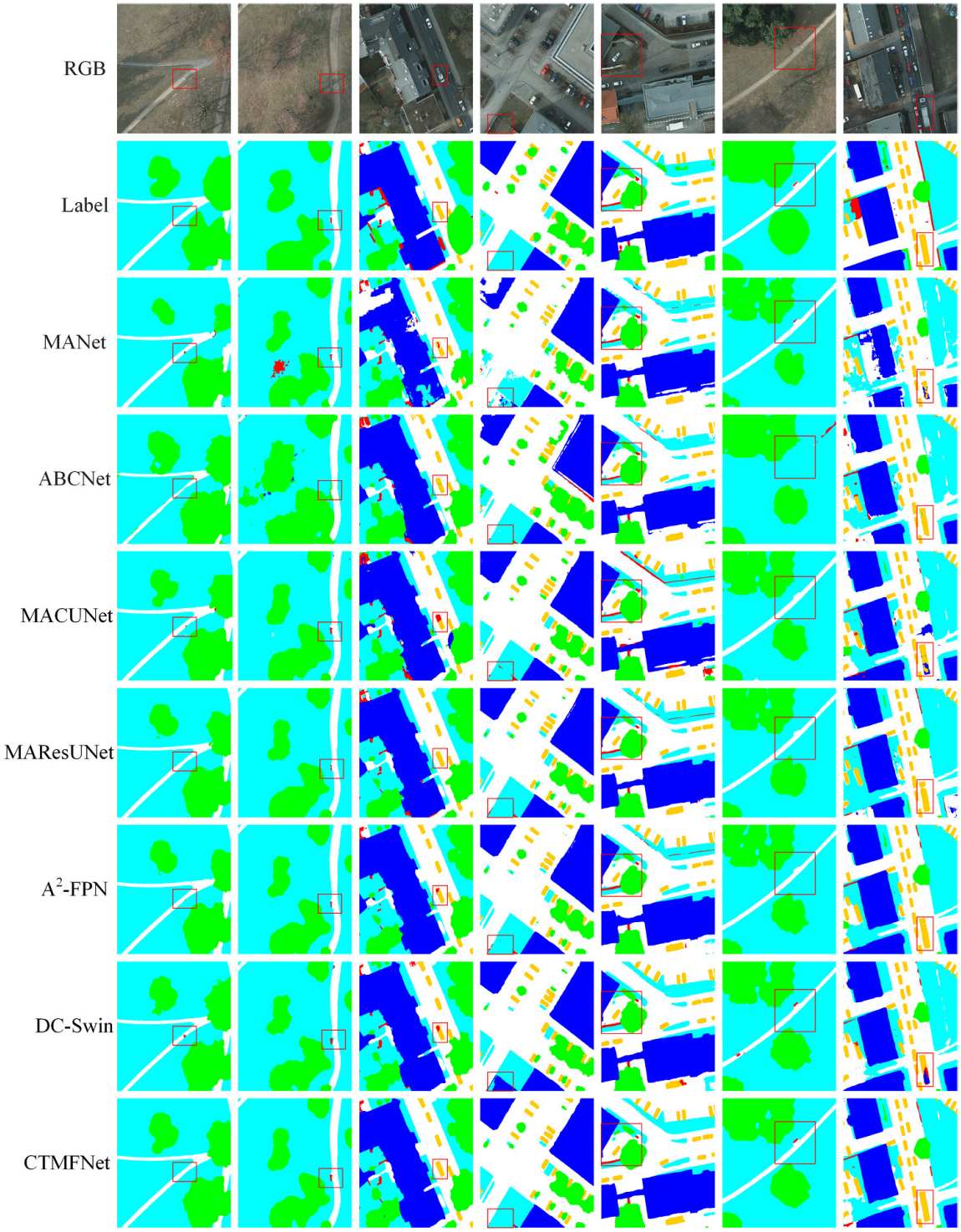



Fig. 10. Qualitative comparisons between our method and other models for the Potsdam test dataset on patches images.

### *C. Experiments With State-of-the-Art Methods on ISPRS Vaihingen*

We show in Table I the quantitative results of our method compared to currently popular methods on the ISPRS Vaihingen dataset. As can be seen from the results in the Table I, we achieved the highest scores for the vast majority of categories. In particular, for the more difficult category "Car," which is a relatively small target object in the image, our method still achieved a dice score of 91.47%. In addition, in terms of average metrics, compared to the suboptimal model, the Mean F1 improved by 1.09%, mIOU by 1.04%, and OA by 0.74%.

To see more visually the difference between our method and the currently popular methods, we show the patch image visualization results for each method in Fig. 8. We also show the segmentation prediction map for the whole high-resolution image and the local prediction details in Fig. 9. As can be seen from the Figs. 8 and 9, CTMFNet achieves better segmentation

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results than currently popular methods. CTMFNet is still able to segment the complete contours of the target objects well when facing target classes with large-scale variability (e.g., building class.), as in the fifth and seventh columns of Figs. 8 and 9. Also, as can be seen from the fourth column in Fig. 8, when faced with a small and dense target of "Car," CTMFNet can segment each car better than other currently popular methods. In summary, CTMFNet uses the transformer to capture global contextual information, expanding the field of perception and generating more fine-grained segmentation prediction maps through semantic communication at multiple scales.

### *D. Experiments With State-of-the-Art Methods on ISPRS Potsdam*

We compare this with currently popular methods on a larger dataset. We show in Table II the quantitative results of our method versus the current popular method on the ISPRS Potsdam dataset. As can be seen from the quantitative results in the table, our method achieved the best results in the vast majority of categories, even getting an F1 score of 96.88% on "Car" In addition, in terms of average metrics, relative to the suboptimal model, our approach is 0.43% higher in Mean F1, 0.75% higher in mIOU, and 0.22% higher in OA.

Similarly, to visualize more clearly the differences between our method and the currently popular methods, we show the patch images and the whole high-resolution image of each method visualized in Figs. 10 and 11. As can be seen in the figure, CTMFNet captures more fine-grained features relative to currently popular methods, as shown in the third and seventh columns in Figs. 10 and 11. We have enhanced the semantic relationships between similar pixels and improved the contextual representation of similar categories by building MDCN. This allows the network to accurately identify categories and segment target boundaries.

### *E. Ablation Experiments*

To verify the soundness and effectiveness of the component design in our approach, we conducted extensive ablation experiments at ISPRS Vaihingen. We propose to design three component models to experiment with the transformer backbone branch, the DAFM, and the MDCN.

*1) Component 1:* We remove the transformer branch from the original method. In DAFM, after reassigning attention weights to local features only, it is transported to the MDCN.

*2) Component 2:* Instead of the DAFM, we use a simple SUM operation to achieve coupling of local and global features.

*3) Component 3:* We replaced the MDCN with the decoding part of UNet to verify the effectiveness of the MDCN.

We show the quantitative results for each component model in Table III. As can be seen from the quantitative results in the table, when deploying the transformer branch significantly improves Mean F1 by 1.28%, mIoU by 1.51%, and OA by 1.22%. This shows that we can extract sufficient global contextual information while retaining fine-grained local features by building a dual-branch backbone. When the DBAM is missing, Mean F1 is reduced by 1.21%, mIoU by 1.39%, and OA by 1.23%. This indicates that a simple summation fusion operation is not the optimal feature fusion solution due to the global and local contextual information. Our DBAM couples local and global features well, creating an effective union mechanism. The results from component three in the table show that the lack of the MDCN makes Mean F1 lower 2.27%, mIoU lower 2.48%, and OA lower 2.13%. This indicates that the MDCN can span the semantic divide between domain scales, establish information exchange between multiple scales and improve the accuracy of segmentation.

We show the qualitative results of the patch image and the whole image for the component model and our method

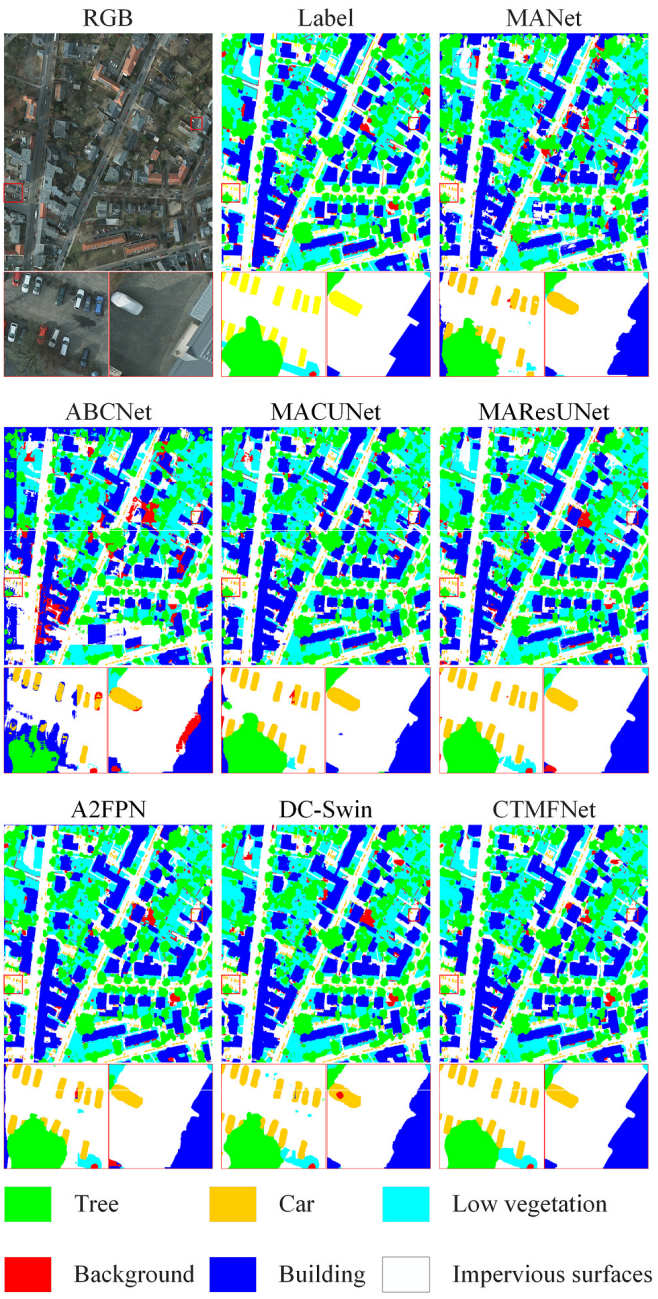



Fig. 11. Qualitative comparisons between our method and other models for the Potsdam test dataset on whole images.

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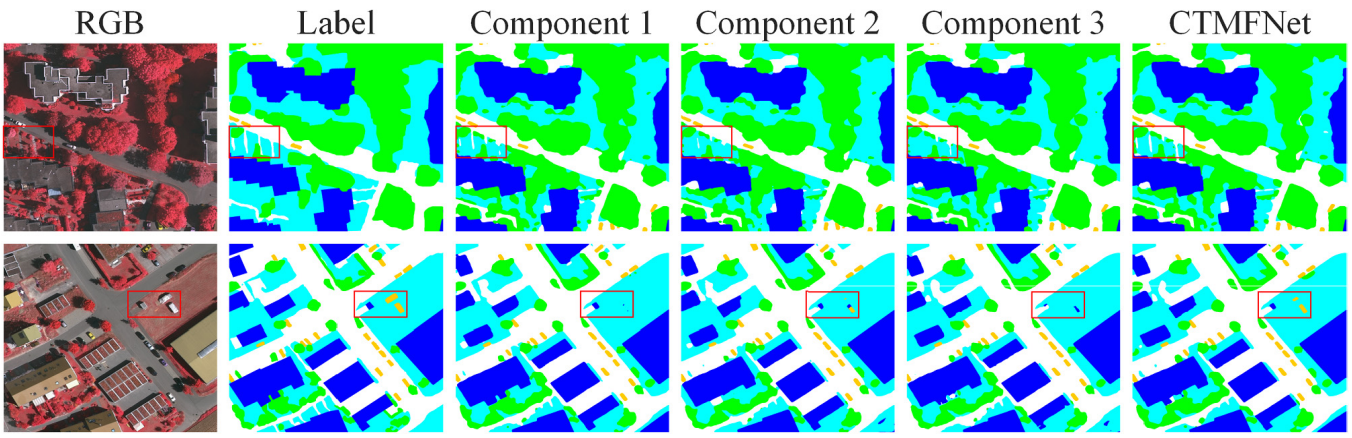



Fig. 12. Qualitative comparisons between our method and other component models for the Vaihingen test dataset on patches images.

TABLE II

QUANTITATIVE COMPARISON OF RESULTS ON THE ISPRS POTSDAM TEST SET WITH STATE-OF-THE-ART MODELS. THE BEST VALUES IN THE COLUMN ARE IN BOLD. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%). INDICATORS FOR THE CATEGORIES ARE F1 SCORES

| Method | Imp.surf. | Building | Lowveg. | Tree | Car | MeanF1 | mIoU | OA |
|---|---|---|---|---|---|---|---|---|
| MANet [12] | 87.46 | 90.66 | 81.96 | 82.81 | 92.76 | 87.13 | 77.45 | 85.05 |
| ABCNet [20] | 87.05 | 88.26 | 78.94 | 84.63 | 93.11 | 86.40 | 76.35 | 84.21 |
| MACU-Net [19] | 87.05 | 88.26 | 78.94 | 84.63 | 93.11 | 86.40 | 76.35 | 84.21 |
| MAResU-Net [29] | 87.05 | 88.26 | 78.94 | 84.63 | 93.11 | 86.40 | 76.35 | 84.21 |
| $A^2$-FPN [30] | 92.85 | 95.84 | 87.17 | 88.76 | 96.13 | 92.15 | 85.66 | 90.68 |
| DC-Swin [22] | **93.26** | 96.86 | 87.74 | 88.68 | 95.50 | 92.41 | 86.10 | 91.16 |
| CTMFNet | 93.22 | **97.12** | **87.87** | **89.36** | **96.61** | **92.84** | **86.85** | **91.38** |

TABLE III

EXPERIMENTAL RESULTS OF THE COMPONENT MODEL. THE BEST VALUES IN THE COLUMN ARE IN BOLD. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%). INDICATORS FOR THE CATEGORIES ARE F1 SCORES

| Method | Imp.surf. | Building | Lowveg. | Tree | Car | MeanF1 | mIoU | OA |
|---|---|---|---|---|---|---|---|---|
| Component 1 | 92.35 | 94.73 | 84.17 | 89.71 | 89.50 | 90.09 | 82.83 | 90.38 |
| Component 2 | 92.19 | 95.02 | 84.22 | 89.64 | 89.71 | 90.16 | 82.95 | 90.37 |
| Component 3 | 91.18 | 94.12 | 83.12 | 88.76 | 88.33 | 89.10 | 81.86 | 89.47 |
| CTMFNet | **93.79** | **96.12** | **85.02** | **90.47** | **91.47** | **91.37** | **84.34** | **91.60** |

in Figs. 12 and 13. As can be seen in Figs. 12 and 13, the CTMFNet lacks the ability to model global contextual information when the transformer branch is missing, and the CNN branch has limitations in capturing long-range feature dependencies. When the network lacks global contextual information, as shown in Fig. 13, the network's response to semantic relationships between similar pixels in the whole image is diminished, causing the network to misidentify categories in the scene. When the network deploys DAFM, it couples local features and global contextual information more effectively than simple fusion operations, enhancing the network' ability to understand multiscale scenes and allowing accurate target outlining. When the network lacks MDCN, CTMFNet lacks sufficient semantic communication in the decoding stage to cross the semantic gap between scales, resulting in some tiny targets in the image being segmented incorrectly.

Finally, our network is analyzed to outperform other methods for several reasons.

1) To compensate for the limitations that exist in CNN itself, we transformer backbone branch to extract global contextual features. Afterward, we designed a DAFM to implement the coupling of local–global features, thus building a powerful union mechanism of CNN and transformer.
2) To span the semantic divide between scales, we build a MDCN as our decoder. The MDCN improves the accuracy of segmentation by up-sampling between scales

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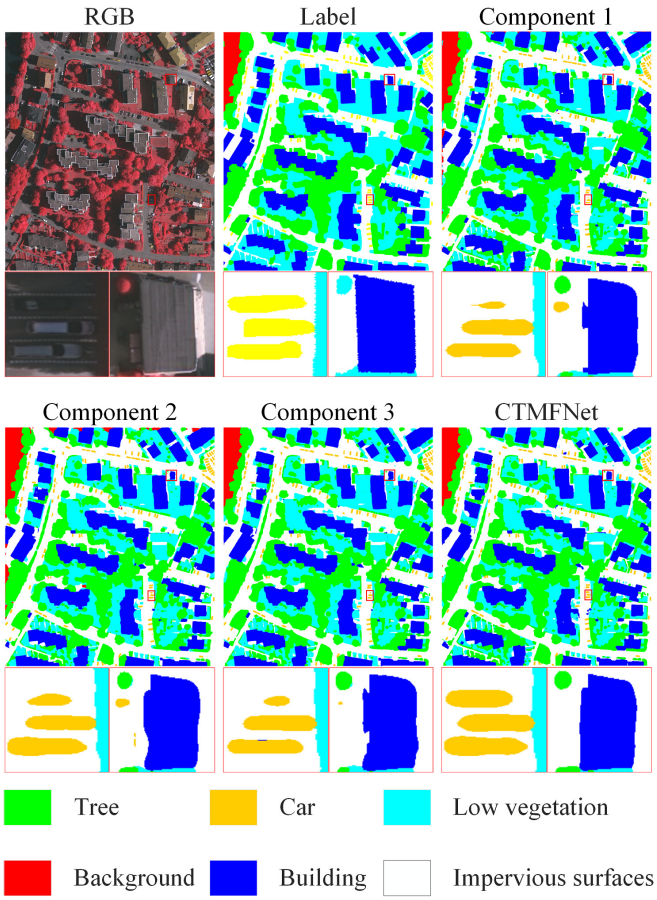



Fig. 13. Qualitative comparisons between our method and other component models for the Vaihingen test dataset on whole images.

and connecting residuals within scales, thus creating a bottom-up, left-to-right information interaction with the co-channel.

## V. Conclusion

In this article, we first propose a new method based on the joint mechanism of CNN and transformer, CTMFNet, to achieve accurate semantic segmentation of remotely sensed urban scene images. Due to the importance of local features and global contextual features in remotely sensed urban scene segmentation, we first set up a dual-branch backbone network of CNN-transformer to extract local detail features and capture the long-range relationships. Second, we designed a DAFM to achieve effective coupling of local features and global contextual information, thus building a more robust CNN and transformer joint mechanism. In the decoding part, we build a multilayer dense connection network, thus establishing a bottom-up channel for the exchange of semantic information. In the decoding part, we build a MDCN, thus establishing a bottom-up channel for the exchange of semantic information. This allows semantic information to flow adequately between multiple scales, crossing the semantic divide between scales. We have demonstrated the effectiveness of our method in remotely sensed urban scene understanding by conducting extensive subjective and objective comparison experiments and ablation experiments on two high-resolution datasets (ISPRS Potsdam and ISPRS Vaihingen).

In addition, our proposed method only performs semantic segmentation of remotely sensed urban scene images and has not been explored for other remote sensing visual tasks (e.g., road segmentation, plot segmentation, etc.). In future work, we intend to further optimize our network structure to accommodate more remote sensing vision tasks. We hope to inspire more researchers to explore the potential and application of the joint mechanism of CNN and transformer in the field of remote sensing.

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